INCH-POUND
MIL-T-83871B(USAF)
15 NOVEMBER 1994
SUPERSEDING
MIL-T-83871A(USAF)
27 JUNE 1979

MILITARY SPECIFICATION

TRANSDUCER, MOTIONAL PICKUP, VIBRATION, VELOCITY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for one type of velocity vibration pickup.

2. APPLICABLE

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods, Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD100	Engineering drawing Practices
MIL-STD-129	Marking For Shipment And Storage

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma Air Logistics Center/TICLA, 3001 Staff Drive, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6695

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MIL-STD-280	Definitions Of Item Levels, Item Exchangeability, Models, And Related Terms
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DOD Material Procedures For Development And Applications Of Packaging Requirements

DRAWINGS

Lockheed-Georgia

3B90071	*Vibration System - Engine*

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM D 1974	Standard Practice For Fabrication Of Fiberboard Shipping Boxes (is not DoD adopted at this time)
ASTM D 5118	Standard Practice For Fabrication Of Fiberboard Shipping Boxes (is not DoD adopted at this time)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order or precedence. In the event of a conflict between the test of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the test of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials used in the manufacture of the pickup shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of Government inspection.

3.3.1 Protective treatment. When materials are used in the construction of the pickup that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met during storage or normal service.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected per MIL-STD-454, requirement 16. This is to include mating surfaces to external equipment.

3.3.4 Prohibitions. Magnesium and magnesium alloys shall not be used.

3.3.5 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3.6 Cadmium. Cadmium shall not be used on or in the pickup without written approval of the preparing activity. Cadmium shall not be used on the exterior of the pickup.

3.4 Design and construction.

3.4.1 General design. The pickup specified herein is a self-generating, electromagnetic velocity transducer and shall be designed to sense engine vibration normal to its longitudinal axis. It shall be a permanent-magnet voltage generator which operates on the seismic mass principle, producing a sinusoidal output for a sinusoidal input.

3.4.1.1 Warmup. The pickup shall not require a warmup period prior to meeting the operational requirements specified herein.

3.4.1.2 Resistive load. The pickup shall be designed to operate into a 10,000 ohm load and maintain all characteristics within specified tolerances.

3.4.2 Construction. The pickup shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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3.4.2.1 Calibration. The pickup shall be constructed such that calibration is accomplished through the magnetization and demagnetization of the permanent magnet using a magnet standardizing coil, RFL Industries Inc P/N HC-7010-16 (or equal), with a magnetizer such as the RFL Industries Inc Model 990 Magnetreater (or equal).

3.4.3 Maintainability. Maintenance of the pickup will consist essentially of cleaning and calibration. The design of the pickup shall be such that requirements for cleaning and calibration are minimized throughout the life of the pickup and neither shall be required more frequent than 500 operating hours. The pickup may be constructed so that the generating device is permanently sealed, in which case, the only maintenance required will be calibration which may be accomplished by treating the permanent magnet material in the generating device.

a. Minimization of the complexity of maintenance tasks (for example: calibration, cleaning, inspections, etc.) by maximum use of simple design and use of standardized equipment or commercial items as approved by the procuring activity.

b. Minimum number and types of tools and test equipment (special and standard)) needed to perform maintenance.

c. The design of the pickup shall be such that requirements for cleaning and calibration are minimized throughout the life of the pickup and neither shall be required more frequent than 500 operating hours.

3.4.4 Reliability. The pickup shall have a minimum acceptable mean-time-between-failure (MTBF) of 2750 operating hours.

3.4.5 Power. The pickup shall require no external power source; the unit shall be self-generating.

3.4.6 Compatibility. The pickup shall be compatible to the engine vibration monitoring system of the C-141A aircraft. The pickup's connector pin wiring, dimensions, and mounting holes shall mate with the aircraft system as specified on Lockheed-Georgia drawing 3B90071.

3.5 Performance. The pickup shall be designed for a normal operating temperature of 800°F. The pickup shall withstand and operate within the limits specified in 3.5.1 through 3.5.7 under any combination of the following conditions:

a. Temperatures - operating temperatures of -65°F to 800°F.

b. Fungus - exposure for 28 days to conditions conducive to the growth of fungus.

c. Humidity - up to 100 percent.

d. Acceleration - acceleration forces up to 23g.

e. Shock - 30g for 11 ±1 milliseconds duration.

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f. Altitude - pressures of 30 inches Hg (sea level) to 3.425 inches Hg ((50,000 feet).

g. Salt fog - exposure to salt fog for a period of 48 hours.

h. Sand and dust - exposure to blowing fine sand and dust.

3.5.1 Insulation resistance. The leakage resistance from the terminals of the electrical connector to the case at 45V dc shall not be less than 50 megohms at normal room temperature; nor less than 0.1 megohm through the temperature range of the pickup.

3.5.2 Statis friction. The maximum acceleration required to overcome static friction in the vertical position shall be 0.25g peak, and 0.62g peak in the horizontal position.

3.5.3 Operating range. The pickup shall operate linearly within a frequency range o 45 to 700 Hz, a maximum displacement of 0.20-inch peak-to-peak, and an acceleration range from 1.1 to 23g's.

3.5.4 Sensitivity. The sensitivity shall be 60 ±2 mv/in/sec peak at 75°F in the vertical position with sinusoidal varying driving force perpendicular to the base at 100 Hz, 2 in/sec peak. This sensitivity shall not vary more than 7% with the pickup installed in any position with the same driving force applied.

3.5.4.1 Resonant frequency. The resonant frequency of the pickup shall be less than 20 Hz.

3.5.5 Linearity. The sensitivity of the pickup shall not vary more than ±5% from the mean sensitivity for either a constant velocity or a constant frequency driving force throughout the area outlined as follows:

a. Frequency - 45 to 700 Hz.

b. Displacement - 0.0009 to 0.032 inches peak-to-peak.

c. Acceleration - 1.1 to 23g's for pickup in vertical position; 1.3 to 23g's for pickup in horizontal position.

d. Force - The vibration pickup must operate at a sustained force of 2.5g's (gravity).

3.5.5.1 High frequency. For frequencies between 700 and 1500 Hz, the sensitivity of the pickup shall be within 60 ±9 mv/in/sec peak.

3.5.6 Temperature error. The sensitivity of the pickup shall not vary more than 0.018%/°F from the reference temperature of 75°F.

3.5.7 Cross coupling. The sensitivity of the pickup to a driving force perpendicular to the sensitive axis shall be less than 2.5% of the sensitivity for the same driving force along the sensitive axis.

3.6 Part numbering. Items supplied to this specification shall be identified as part number M83871-1.

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3.6.1 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be interchangeable as defined in MIL-STD-280. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number an changes thereto.

3.7 Connector. The connector shall be configured to receive the existing coupling system of the C-141A aircraft as specified on Lockheed-Georgia drawing 3B90071.

3.8 Dimensions. The dimensions of the pickup shall be within the maximum limits as specified in Lockheed-Georgia drawing 3B90071.

3.9 Weight. The weight of the pickup shall not exceed 4 ounces.

3.10 Sealing. The case shall provide a sealed enclosure for all mechanical and electrical components of the pickup. The sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any environment to which the pickup may be subjected. It shall be sealed sufficiently to meet the requirements and pass the tests herein.

3.11 Identification of product. Each pickup shall be marked for identification in accordance with MIL-STD-454, requirement 67.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.12.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions and tolerances affect he interchangeability, operation, or performance of the pickup, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirement, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1.

4.3.1 Standard atmospheric conditions. Whenever he pressure and temperature existing at the time of the test are not specified, it is understood that the test will be made at normal room conditions and a temperature of approximately 75°F. Should the temperature differ appreciably from the above value, proper allowances shall be made.

4.3.2 Standard operating position. Unless otherwise specified, the pickup shall be tested with its sensitive axis vertical, base down, and the driving force along the sensitive axis.

4.3.3 Test sequence. Unless otherwise specified, the tests shall be conducted in the sequence listed in 4.6.

4.4 First article inspection. The test samples shall consist of 15 pickups representative of the production equipment. Three of the pickups shall be subjected to all the tests of 4.6 except Reliability, 4.6.9. The other 12 pickups shall be subjected only to the individual tests, 4.5.1, and the reliability test.

4.5 Quality conformance inspection. The quality conformance tests shall be of two types:

a. Individual tests.

b. Sampling plans.

4.5.1 Individual tests. Each pickup shall be subjected to the following tests as described under 4.6.

a. Examination of product	See 4.6.1
b. Stabilization	See 4.6.2
c. Insulation resistance	See 4.6.3
d. Sensitivity	See 4.6.4
e. Resonant frequency	See 4.6.5
f. Static friction	See 4.6.6
g. Cross coupling	See 4.6.7

4.5.2 Sampling plans.

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4.5.2.1 Sampling plan A. One pickup shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests	See 4.5.1
b. High temperature	See 4.6.8.1
c. Low temperature	See 4.6.8.2
d. Leakage	See 4.6.8.3

4.5.2.2 Sampling plan B. Under otherwise specified by the procuring activity, 3 pickups, selected at random from the first 15 produced on contract or order, shall be subjected to the following tests as described under 4.6.

a. Sampling plan A	See 4.5.2.1
b. Sand and dust	See 4.6.8.4
c. Salt fog	See 4.6.8.7
d. Shock	See 4.6.8.8
e. Temperature, humidity, altitude	See 4.6.8.9

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.4 Individual tests may continue. For operational reasons, individual tests may continue pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.6 Test methods.

4.6.1 Examination of product. The pickup shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, weigh, finishes, dimensions, and any visible defects.

4.6.2 Stabilization. The pickup shall be subjected to a 12-hour stabilization period. During stabilization the pickup shall be vibrated such that the magnetic bobbin very nearly travels its full distance. The pickup shall be in the horizontal position and rotated at 4 rpm about its sensitive axis throughout the stabilization process.

4.6.3 Insulation resistance. The leakage resistance from each terminal of the electrical connector to the case shall be checked. A voltage of 45V dc shall be used and the resistance shall be greater than 50 megohms.

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4.6.4 Sensitivity.

4.6.4.1 Calibration. The pickup shall be mounted on a vibration table. A driving force of 100 Hz ad 2.0 in/sec peak shall be applied along the sensitive axis. The output of the pickup shall be monitored for a sensitivity of 60 ±2 mv/in/sec peak. If necessary, the pickup shall be calibrated through demagnetization for the above sensitivity. NOTE: For the above test and for all succeeding tests which involve the measurement of the pickup output, the output terminals of the pickup must be connected across 10K resistor.

4.6.4.2 Output waveshape. The pickup shall be mounted on a shaker table, its output monitored by an oscilloscope. The shaker table shall be set up for a 100 Hz, 2 in/sec peak driving force. After the shaker table has stabilized, the output of the pickup shall be smooth, undistorted sinusoidal waveshape.

4.6.4.3 Linearity. With the pickup mounted on a vibration table in its standard operating position, 4.3.2, the sensitivity of the pickup shall remain within ±5% of the mean sensitivity for the following drive forces without further adjustments to the pickup:

FREQUENCY (Hz)	VELOCITY (in/sec peak)
45	2.0
60	2.0
100	2.0
200	2.0
300	2.0
400	2.0
500	2.0
600	2.0
700	2.0
100	0.707
100	1.0
100	5.0
100	10.0

4.6.4.3.1 Position linearity. The pickup shall be mounted on a vibration table with a driving force of 100 Hz and 2 inc/sec peak applied along the sensitive axis. The table shall be rotated 360° and measurements made of the sensitivity at 45 degree intervals. Each of these measurements shall be within ±7 percent of the sensitivity obtained in 4.6.4.1. Rotation of the vibration table 360° may be simulated by mounting the pickup upside down on a fabricated metal fixture and rotating the table only 90°.

4.6.4.4 High frequency. With the pickup mounted in the standard operating position, the sensitivity shall remain within 60 ±9 mv/in/sec peak for the following drive forces without further adjustments to the pickup:

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FREQUENCY (Hz)	VELOCITY (in/sec peak)
1000	0.5
1200	0.5
1400	0.5
1500	0.5

4.6.5 Resonant frequency. The pickup shall be installed on a vibration table in its standard operating position, 4.3.2. The output of the pickup shall be connected to the vertical input of an oscilloscope. The output of the vibration table signal generator shall be connected to the horizontal input of the oscilloscope. The vibration table shall be set to vibrate at a frequency of approximately 20 Hz and a velocity of 3 in/sec peak. The frequency of vibration shall be varied slowly from 20 Hz until a 90° phase shift is observed on the oscilloscope. The frequency at which the phase shift takes place shall be less than 20 Hz.

4.6.6 Static friction. With the pickup vertically mounted on a vibration table set for 100 Hz, the velocity shall be increased from 0 in/sec until a noticeable output from the pickup is detected. The velocity shall be less than 0.25 in/sec peak. This procedure shall be repeated with the pickup mounted in the horizontal position and the driving force still along the sensitive axis. The resulting velocity shall be less than 0.62 in/sec peak.

4.6.7 Cross coupling. The pickup shall be mounted horizontally and a driving force of 100 Hz, 2 in/sec peak applied perpendicular to the sensitive axis. The output response shall be less than 2.5% of that obtained in 4.6.4.1.

4.6.8 Environmental tests. Each of the following environmental tests shall use as pretest and past test data; sensitivity, insulation resistance, resonant frequency and static friction. Where the pickup is to be operated during tests, these same parameters shall be used with proper consideration given for temperature. Failure to meet any of the requirements specified herein shall constitute failure of the pickup. Under no conditions shall the pickup require a warmup period after the driving force has stabilized to meet the specification.

4.6.8.1 High temperature. The pickup shall be tested in accordance with MIL-STD-810C, method 501.1, procedure II. The highest operating temperature for the pickup shall be 800°F. Insulation resistance (4.6.3) of the pickup after the temperature, has stabilized at 800°F shall be not less than 0.1 megohm.

4.6.8.2 Low temperature. The pickup shall be tested in accordance with MIL-STD-810C, method 502.1, procedure I, except the storage temperature shall be the standard ambient temperature. The lowest operating temperature shall be -65°F. Insulation resistance (4.6.3) of the pickup after the temperature has stabilized at -65°F shall be not less than 0.1 megohm.

4.6.8.3 Leakage. The pickup shall be tested in accordance with MIL-STD-810C, method 512.1, procedure III. The absolute pressure of the air above the test liquid shall be reduced to 1 inch of mercury absolute.

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4.6.8.4 Sand and dust. The pickup shall be tested in accordance with MIL-STD-810C, method 510.1, procedure I, except that the pickup shall be operated at 200 Hz with a 2 in/sec peak driving force during the two 6-hour periods. Also, the pickup shall be rotated such that all sealed openings are exposed equally.

4.6.8.5 Fungus. The pickup shall be tested in accordance with MIL-STD-810C, method 508.1, procedure I. The pickup need not be operating during the test.

4.6.8.6 Humidity. The pickup shall be tested in accordance with MIL-STD-810C, method 507.1, procedure I. The pickup need not be operated prior to step 7 other than for pretest data.

4.6.8.7 Salt fog. The pickup shall be tested in accordance with MIL-STD-810C, method 509.1, procedure I.

4.6.8.8 Shock. The pickup shall be tested in accordance with MIL-STD-810C, method 516.2, procedure I using FIGURE 516.2-2, half sine shock pulse configuration, with a peak acceleration of 30g's and a duration of 11 milliseconds. The pickup need not be operating during the test.

4.6.9 Reliability testing. Twelve pickups shall be subjected to a reliability test in accordance with MIL-STD-781. The temperature during testing shall remain at a constant 800°F and the pickup shall be vibrated at a frequency sweep of 45 to 200 Hz with an amplitude of 7.0 mils peak-to-peak. The frequency cycling rate shall be approximately 6 times per hour. Accept-reject criteria shall be in accordance with test plan IV of MIL-STD-781.

4.6.9.1 Performance testing. Prior to the start of the reliability test, the individual tests, 4.5.1, shall be performed on each pickup to establish a performance record. Thereafter, the individual tests (except for stabilization and cross coupling) shall be performed on each pickup at intervals of 72 operating hours or less.

4.6.9.2 Relevancy of failures. The inability of the pickup to provide correct outputs within the accuracy requirements of the individual tests shall constitute a failure. No adjustment of the pickup shall be necessary to meet these accuracy requirements before the pickup has accumulated at least 500 hours operating time from its last adjustment. All failures shall be classified as relevant failures unless approved for exclusion by the procuring activity. All failures shall be investigated fully to ascertain if the failure is a repeatable phenomenon and has occurred in the pickup and not in the test equipment.

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

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5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be large enough to allow for the application of cushioning/wrap material, of sufficient density and thickness, to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for the application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to ASTM D 5118, weather-resistant, unless less otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Exterior container closure shall be IAW ASTM D 1974. Sealing Method "A" procedure and the contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contracting activity and MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The velocity vibration pickup covered by this specification is intended to be used on the C-141A aircraft's TF-33P jet engine to convert the mechanical vibrations to electrical signals which are to be transmitted to an indicator.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Levels of packaging and packing required, special shipment marking, and reinspection date marking.

c. When sampling plan B is not be conducted.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 First article. When first article inspection is required the contracting officer should provide specific guidance to offerors whether the item(s) should be a reproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instruction in acquisition document regarding arrangement for examinations, approval of first article test results, and for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (keyword listing).

Motional Pickup
Transducer
700 HZ
23G

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing Activity:
AIR FORCE -71

Project Number
6620-F542

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-83871B(USAF)	2. DOCUMENT DATE (YYMMDD) 94 11 15

3. DOCUMENT TITLE
TRANSDUCER,MOTIONAL PICKUP,VIBRATION VELOCITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial	(2) AUTOVON
OC-ALC/TICLA		336-5960
c. ADDRESS *(Include Zip Code)* 3001 Staff Dr Tinker AFB OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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